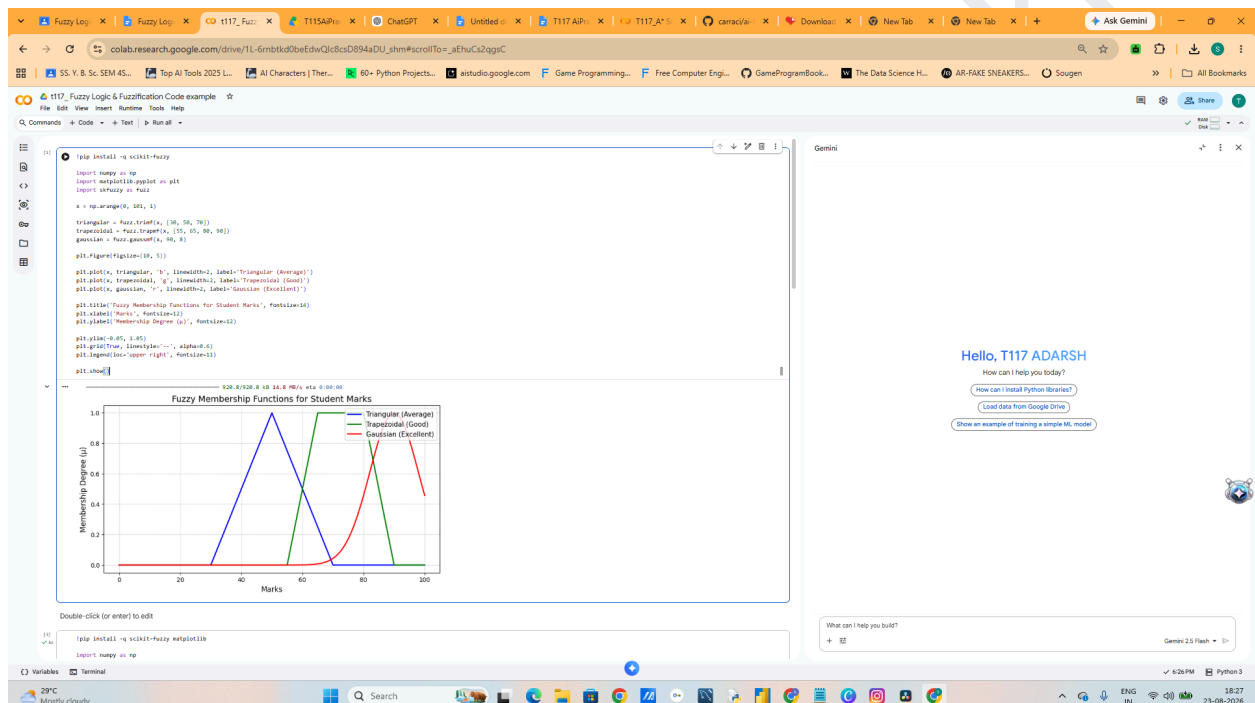


SHETH L.U.J AND SIR M.V. COLLEGE
SUBJECT NAME: Artificial Intelligence

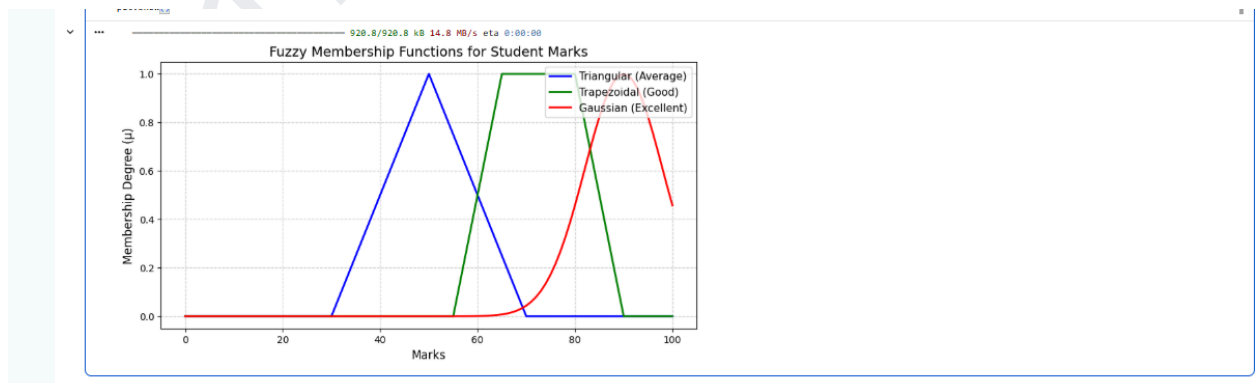
Practical 4: Fuzzy Logic & Fuzzification Code example

AIM : Fuzzy Logic & Fuzzification Code example

Code :



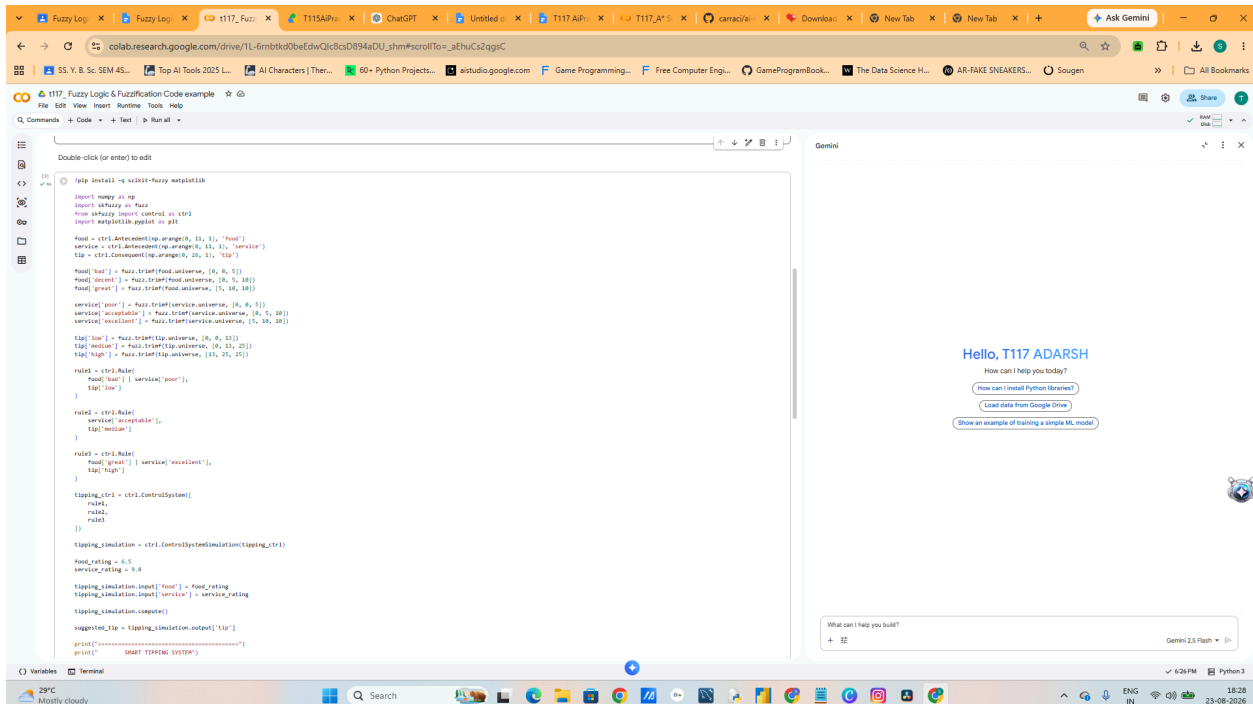
Output :



Name :Name: Adarsh Shukla
Roll no :no.: T117

SHETH L.U.J AND SIR M.V. COLLEGE
SUBJECT NAME: Artificial Intelligence

Practical 4: Fuzzy Logic & Fuzzification Code example



```
import numpy as np
import sys
from fuzzy import control as ctrl
import matplotlib.pyplot as plt

food = ctrl.Antecedent(np.arange(0, 11, 1), 'food')
service = ctrl.Antecedent(np.arange(0, 11, 1), 'service')
tip = ctrl.Consequent(np.arange(0, 26, 1), 'tip')

food['low'] = fuzz.triangular(food.universe, (0, 0, 5))
food['medium'] = fuzz.triangular(food.universe, (0, 5, 10))
food['high'] = fuzz.triangular(food.universe, (5, 10, 10))

service['low'] = fuzz.triangular(service.universe, (0, 0, 5))
service['medium'] = fuzz.triangular(service.universe, (0, 5, 10))
service['high'] = fuzz.triangular(service.universe, (5, 10, 10))

tip['low'] = fuzz.triangular(tip.universe, (0, 0, 13))
tip['medium'] = fuzz.triangular(tip.universe, (0, 13, 25))
tip['high'] = fuzz.triangular(tip.universe, (13, 25, 25))

rule1 = ctrl.Rule()
rule1['food'] = service['low'],
tip['low']

rule2 = ctrl.Rule()
rule2['service'] = service['acceptable'],
tip['medium']

rule3 = ctrl.Rule()
rule3['great'] = service['excellent'],
tip['high']

tipping_ctrl = ctrl.ControlSystem([
    rule1,
    rule2,
    rule3
])

tipping_simulation = ctrl.ControlSystemSimulation(tipping_ctrl)

food_rating = 6.5
service_rating = 9.8

tipping_simulation.input['food'] = food_rating
tipping_simulation.input['service'] = service_rating

tipping_simulation.compute()

suggested_tip = tipping_simulation.output['tip']

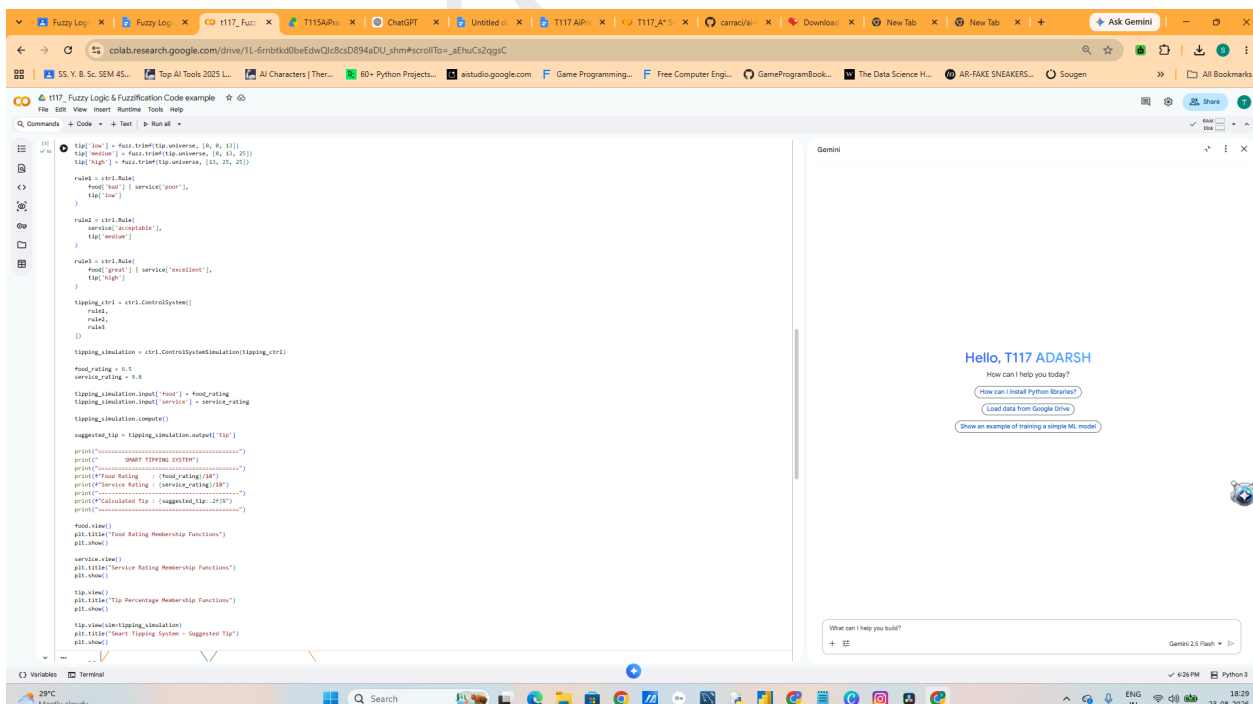
print("=====")
print("SMART TIPPING SYSTEM")
print("=====")
print("Food Rating : (food_rating/10)")
print("Service Rating : (service_rating/10)")
print("Calculated Tip : (suggested_tip/25)")
print("=====")

food.view()
plt.title("Food Rating Membership Functions")
plt.show()

service.view()
plt.title("Service Rating Membership Functions")
plt.show()

tip.view()
plt.title("Tip Percentage Membership Functions")
plt.show()

tip.view(sim=tipping_simulation)
plt.title("Smart Tipping System - Suggested Tip")
plt.show()
```



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plt.title("Smart Tipping System - Suggested Tip")
plt.show()
```

Name :Name: Adarsh Shukla
Roll no :no.: T117

Practical 4: Fuzzy Logic & Fuzzification Code example

Output :

